

PRATIMAI · AI HANDOUTS

The Logic Under the Magic

My case file for the maths that makes machines learn — and the reasons they fail

CLASS 9 · LEVEL 4 — REASON

21 PAPER MISSIONS · 5 CHAPTERS · NO SCREEN NEEDED

PRATIMA AI Curriculum Team

How do I know? · What is the average hiding? · What is the trick here?

THIS CASE FILE BELONGS TO

name · class · roll number

CASE OPENED ON ★ CASE CLOSED ON

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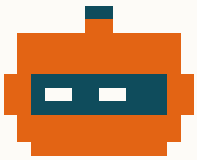
Every activity in this book is **unplugged**: no task requires a device, an account, a photo, a voice recording or any personal information. All examples use invented people and places. The Class 9 material is built on the last official CBSE Class IX AI curriculum structure (2025-26), ICSE research-and-AI units and NCERT's computational-thinking direction — with programming deliberately kept out: every mechanism here runs on paper first.

THE LICENCE IN PLAIN WORDS

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Set in **Nunito** and **Baloo 2** on warm paper stock. Both typefaces are used under the SIL Open Font Licence. The book's diagrams are printed as crisp, high-resolution figures, and every page was reviewed individually before this edition went to press.



PRATIMAI · AI HANDOUTS

Acknowledgments

To the Classes 6, 7 and 8 investigators whose case files walked the whole road with us — from spotting learning machines to planning them on paper. Your notebooks convinced us that school students can run real statistics, real probability and real graph criticism when the numbers belong to problems they care about. This book finally hands you the engine room: the mathematics under the machine.

To the maths teachers who read every mission and said "yes — but make the outlier crueler, make the dice honest, make the best-fit line earn itself": thank you. You protected the book from fake maths and from the greater sin, boring maths. And to the adults who will be asked "is the AI's average the same as MY average?" at the dinner table: by the last page, your student will answer you with a worked example.

This series was built brick by brick by three teams working as one. The **AI research team** checked every explanation against how machines really learn, so nothing in this book is a myth wearing a friendly font. The **academic team** walked every chapter beside the school syllabus, so each game quietly trains a skill that examinations will ask for later. And the **child-psychology team** shaped the pace of every page: short missions for young attention, quick wins that open every chapter, confidence circles that make honesty feel safe, and space to write on every task — because an idea worked by the hand is remembered by the mind. Alongside them, curriculum reviewers guarded the "unplugged" promise: not one mission in this series asks for a phone, an account or a password.

Thanks go as well to the designers of Nunito and Baloo 2, whose friendly letterforms keep pages of hard reasoning feeling light. Finally, to the AI Analyst holding this book: every dice roll you have not rolled yet, every line you have not drawn yet and every graph you have not yet caught lying is the real content of this case file. We only set the stage. You bring the evidence.

21 PAPER MISSIONS · 20 WORDS TO KEEP · 5 CHAPTERS · 1 BADGE EARNED

How to Use This Handbook

Welcome back, Analyst. Three years of evidence sit behind you: in Class 6 you **spotted** learning machines, in Class 7 you **trained and tested** them, in Class 8 you **planned and defended** them. Now comes the question every honest engineer eventually faces: **why does any of it work?** This year you go under the hood — and the hood opens with mathematics, not code. Averages that mislead, probability you can roll on a desk, lines you draw through real dots, neighbours you classify with a ruler: every mechanism that makes AI possible, run by hand. And every place the same mathematics lets machines — and their marketing — fail. You will still need no computer. A pencil, a ruler, a pair of dice and a suspicious mind are the complete toolkit.

WHY THIS BOOK WORKS THE WAY IT DOES

This handbook is built around how teenage brains actually learn. Every chapter opens with a quick win — motivation first, theory second. You **do** before you **define**: each idea is met through an experience before it is named, because memory hooks onto action. The confidence circles train metacognition — knowing how sure you are is a skill, and honest rating makes it stronger. Tasks are time-boxed because attention works in sprints; ideas return across chapters on purpose, because a second meeting is what turns exposure into understanding; and every task leaves space to write, because producing your own answer beats re-reading every time. Nothing here can be failed: a wrong answer you go on to check is a clue found early.

1 READ

Read the mission twice. Underline anything that puzzles you — puzzles are clues.

2 DO

Do the task with your partner, group, class or family, exactly as the labels say.

3 WRITE

Write what you computed, then shade your confidence circles honestly.

The work-mode labels

ALONE	your own case — you investigate by yourself
PAIR	two analysts — share one book, take turns reading codes aloud
GROUP	three to four analysts — decide roles before you start
CLASS	the whole class plays together, led by your teacher
HOME	a mission for home — show an adult what you discovered

HOW TO READ A MISSION

T9-10 Mission title — **ALONE · 15 MIN**

■ **QUICK WIN** marks the easier missions; ■ **HANDS-ON LAB** marks the working labs — missions where you **build, measure, play or draw**.

Every mission has a code like **T9-10** — **T9** means this Class 9 handout, and **10** is the mission number, so your teacher can say “open Task T9-13”. The **ALONE** label tells you who to work with, **15 MIN** is your time limit, and **QUICK WIN** marks an easy first success. This book’s missions ask for **working, not just answers** — show the division, draw the line, mark the outlier. In mathematics, the working IS the reasoning.

THE QUESTION HABIT — YOUR SUPERPOWER

Your five questions — **"How do I know?"**, **"What is missing?"**, **"Who made this?"**, **"Who is missing?"**, **"What is the trick here?"** — meet their sharpest field of play yet: numbers themselves. This year add the analyst's suffix to every statistic you meet: **"compared to WHAT, over WHICH period, leaving out WHOM?"** An average without its context is a magic trick. Your job, from page one, is to be the person who asks how the trick is done.

DRILL THE QUESTIONS — ON A CLAIM WITH NUMBERS

A news graphic says: "Students who use AI tutors score 20% higher." Run the analyst's questions and write one honest line each:

1. **How do I know?** — 20% of what, compared with whom, over how long?

2. **What is missing?** — Which average, and what could an outlier be doing here?

3. **Who made this?** — Who sells the tutor? Re-weight the claim.

4. **Correlation or cause?** — Name the hidden factor a careful analyst would check.

A Note for Adults

This handout builds **AI understanding** — how machines learn from data and how humans should judge them — through 21 paper-and-pencil missions. It is fully **unplugged**: no task requires a device, an account, a photo, a voice recording or any personal information. This year the big ideas are the project cycle with 4Ws problem scoping, stakeholders and system maps; data literacy and graph criticism; the statistics and probability that power learning (mean, median, mode, spread, likelihood, line of best fit, nearest neighbours); generative AI at concept level — how text generators guess and why they invent; and ethical reasoning with SDG-linked project briefs. The mathematics stays inside the Class 9 syllabus — statistics, probability, coordinate geometry — but every formula is earned through an experience first. The best help you can give is to ask for the **working**, not just the answer, and to treat confidence circles as honest thinking, not scores. A separate teacher pack carries answer notes and rubrics; this book deliberately never prints solutions beside a task — only short hints at the very back.

Four ways to help — without giving answers

1 Ask for the working

When your analyst is stuck, ask: **“show me the calculation you trust least”** — the weakest step is where the real lesson lives.

2 Praise the scepticism

Swap “correct!” for **“you asked what the average was hiding”** or **“you checked a second source before believing.”** Scepticism is a skill; name it when you see it.

3 Let the dice be rolled

Probability becomes real the first time a prediction fails. Let the coins be flipped, the tally be messy, the surprise be genuine.

4 Connect to the news

When a headline quotes a number, ask **“what would YOU need to know before trusting it?”** This book trains exactly that reflex.

IF YOUR CHILD ASKS…

“Is the maths in this book the REAL maths AI uses?” — Say: yes, at its foundation. Averages, spread, probability, best-fit lines and nearest-neighbour distance are genuinely how learning systems reason; the classroom versions are run by hand instead of by silicon. **“Why does the chatbot invent things?”** — Say: it predicts likely next words from patterns; likely is not the same as true. This book builds a next-word table by hand so your child can see the invention happen. **“Will this help in exams?”** — Say: the statistics, probability and graph-reading are syllabus mathematics — but the habit of asking what a number hides is worth more than any single mark.

SIGNS OF ANALYST GROWTH WORTH PRAISING

Watch for these quiet changes — they matter more than any correct answer: your child **shows the working** without being asked · they say **“one number can’t tell the whole story”** and mean it · they **roll the dice twice** when a result surprises them · they ask **“compared to what?”** at advertisements and headlines · they defend a claim with **evidence and its limit** in the same sentence. When you spot one, name it: “That is exactly what analysts do.”

My AI Journey Map

THE FIVE STRANDS YOU WILL MEET AT EVERY STOP

U Understand AI
How do rule-based and learning systems differ — and where do AI, ML and DL sit?

D Data
Can I judge a dataset's quality — and catch a graph that lies?

L Learn
Which mathematics makes learning possible: averages, probability, best-fit lines?

W World
How does generative AI really work — and how do I verify it?

R Responsibility
Who gains, who loses, who decides — and how do I reason about it?

THINKING-TOOLS THREAD

runs through every stop: **decompose** the problem · **compute before you claim** · **compare competing explanations** · **trace the procedure, step by step**

THE FIVE STOPS ON YOUR CASE

1 **Reflect, Scope, Plan**
4Ws · stakeholders · system maps

2 **Data Literacy**
types · quality · misleading graphs

3 **The Maths Behind AI**
averages · probability · best fit · kNN

4 **Generative AI & Ethics**
next-word guessing · deepfakes · verify

5 **SDG-linked AI Brief**
brief · ethics panel · flowchart trace

BY THE END OF THIS BOOK, I CAN…

- [ ] tell AI, machine learning and deep learning apart — and rule-based from learning-based systems
- [ ] run the project cycle with 4Ws problem scoping, stakeholders and a system map of data features
- [ ] classify data as qualitative or quantitative, structured or unstructured — and judge its quality and sources
- [ ] read a graph critically and name its trick: truncated axes, cherry-picking, misleading pie shares
- [ ] compute and interpret mean, median, mode and spread — and explain when an average misleads
- [ ] compute simple probabilities and connect them to a machine's "confidence"
- [ ] fit a line by eye through real data, predict with it, and explain why correlation is not causation
- [ ] classify a new point by nearest labelled neighbours, with distances as evidence
- [ ] explain generative AI conceptually — and run a verification routine on anything it produces
- [ ] argue an AI case from several stakeholders' views, with an ethics checklist
- [ ] link an AI idea to an SDG with a data plan and an ethics note
- [ ] express a decision procedure as a flowchart and trace it by hand, step by step

THREE NUMBERS I WILL DISTRUST THIS WEEK

An analyst's year starts with suspicion, not cynicism. List three numbers the world will show you this week — a fee, a claim, a score — and name the question you will ask each:

1. _____ h(4pt)
my question → _____ h(4pt)
2. _____ h(4pt)
my question → _____ h(4pt)
3. _____ h(4pt)
my question → _____ h(4pt)

DAY 1 My starting line — a pledge to the evidence

2 MIN

Fill this in before Chapter 1. The last page will ask for the year's evidence next to these lines — vague pledges fail audits.

One number in daily life whose average I now intend to distrust on sight:

One claim about AI I want to be able to **prove or refute** by June:

My honesty clause — how I will record results that contradict what I hoped for:

Signed: _____ h(6pt) Date: _____ h(6pt)

FIRST IMPRESSIONS — THE ANALYST'S FLIP TEST

Flip through this book for two minutes, like an auditor meeting a new client. Three things I noticed that I intend to question later:

1. _____ h(4pt)
2. _____ h(4pt)
3. _____ h(4pt)

CARD The analyst's five questions — my pocket copy

1. **How do I know?** — what was measured, how, on whom?
2. **What is missing?** — which average, and what does it throw away?
3. **Who made this?** — who gains if I share it without checking?
4. **Who is missing?** — whose outcome was never counted?
5. **Correlation or cause?** — what else could explain the pattern?

The claim I intend to check first — and the question that will crack it:
_____ *h(4pt)*

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The boxes in this book

You will meet these boxes on every case. Each one wants something different from you — collect them all.

MISSION

A paper mission with a code like **T6-01**. The label tells you who you work with and how many minutes you get. Do it — then write. Writing **is** the mission.

WORD POWER

A word worth keeping in your analyst kit. Read it three times today — words are the tools of thinking.

SELF-CHECK

Skills to grade yourself on. Shade a confidence circle honestly — honest circles make you learn faster.

THINK & INK

A question with lines. There is no single right answer here — your **reasons** are the answer.

Also inside: **MYTH BUSTER** busts a wrong idea many people believe · **ANALYST'S NOTE** hands you a professional's secret · **HOME LINK** (dashed) is a mission to take home and teach an adult.

1. Reflect, Scope, Plan

GUIDING QUESTION

Why does AI work — and when does it fail?

The engine room opens. Before any mathematics, you need the map: what kind of system are we even talking about? This chapter separates rule-based machines from learning-based ones, places AI, machine learning and deep learning inside each other like nested boxes, and then rebuilds your planning toolkit at analyst level — the 4Ws canvas, stakeholder maps, and a system map that names every data feature a machine will actually see. Every AI failure you will ever study begins with a skip in one of these steps.

MISSIONS T9-01 – T9-04
OUTCOMES 9.U1 · 9.L1 · 9.R1 **STRANDS** U L

Links: Maths — data & coordinates · Social Science — stakeholders · English — precise definitions

In this chapter

1.1 Three nested boxes: AI, ML, DL 11
1.2 Stakeholders: everyone the machine touches 13
1.3 The system map: name every feature the machine sees 14
1.4 Rule vs learn: the showdown 16

TERMS YOU'LL ANALYSE WITH

deep learning · rule-based system · learning-based system · problem scoping · system map

WARM-UP · BEFORE YOU READ ALONE · 2 MIN

A chatbot writes you a poem about the monsoon. Write your honest one-line answer: did it LOOK UP that poem, or something else happened? Keep your answer — you will test it in Chapter 4.

ANALYST'S TOOLKIT — YOU WILL NEED: pencil · ruler · sticky notes · your Class 8 proposal for re-reading

1.1 Three nested boxes: AI, ML, DL

Time to make the vocabulary honest. **Artificial intelligence** is the whole field: any machine doing things we used to need human judgement for — including old-fashioned rule-followers. **Machine learning** sits inside it: systems that improve from examples instead of from hand-written rules. **Deep learning** sits inside machine learning: learning systems built from many layered pieces, powerful enough to find patterns in raw speech, pixels and text — and hungry enough to need enormous data. So every deep-learning system is machine learning, and every machine-learning system is AI — but plenty of AI is neither. The lift in your building is AI by the widest definition; it will never learn anything.

AT A GLANCE · THE NESTED BOXES — AND THE GREAT DIVIDE

The system	Inside which box?	Learns from examples?	One honest limit
a lift with floor buttons	AI only	never — every rule was typed	it will never improve
a spam filter	ML (inside AI)	yes — from labelled mail	new tricks fool it
a voice assistant	DL (inside ML)	yes — from oceans of speech	accents it rarely heard
a chatbot that writes essays	DL (inside ML)	yes — from oceans of text	it predicts, it never verifies

*The great divide cuts ACROSS the boxes: **rule-based** (a person wrote every step) versus **learning-based** (the system found the pattern in examples). Ask it of every machine you meet this year.*

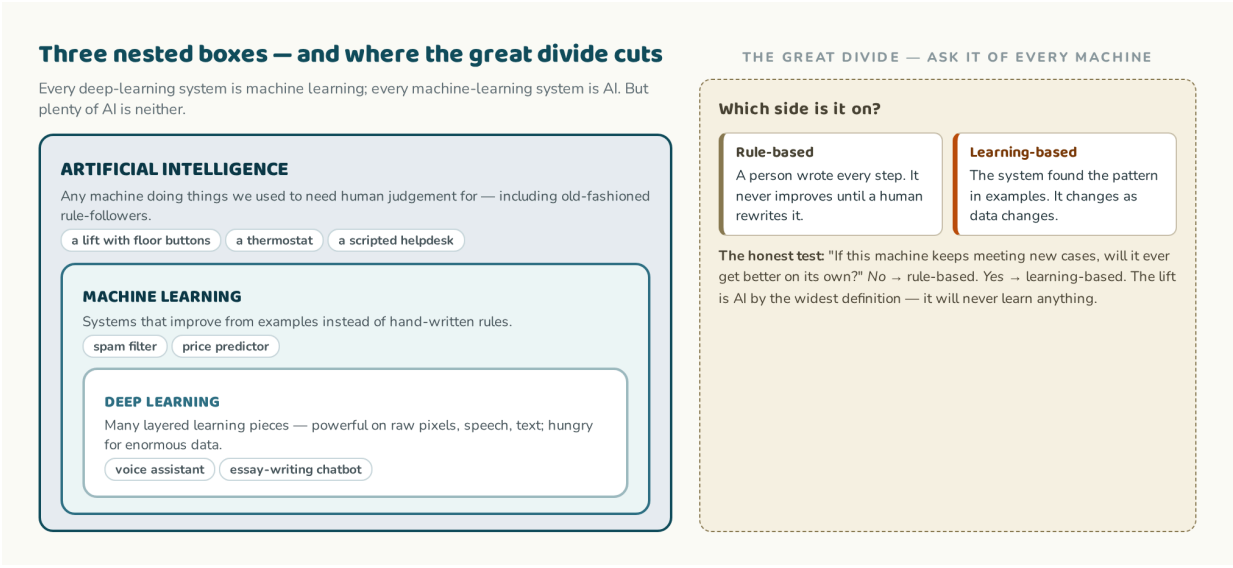


FIG The nested boxes as a picture, with the great divide beside them — keep this image for the checkpoint.

T9-01 4Ws Problem Canvas

GROUP · 20 MIN

Part A of the cycle you will run all year: **scoping**. Problem scoping is the craft of pinning down exactly which problem to solve, for whom and where — before any solution is discussed. The canvas has four quadrants. Fill it for a real problem in your school or neighbourhood.

W1 · WHO has the problem? *The people who feel it daily — name them specifically*

W2 · WHAT is the problem? *One sentence — an outcome to change, not a solution*

W3 · WHERE does it appear? *The exact places, times, seasons*

W4 · WHY is it worth solving? *The cost of doing nothing — to each group*

Our problem statement from the four Ws (one sentence):

The trap we nearly fell into (a solution we wanted to discuss before scoping):

WORD POWER · 1

Definition: **problem scoping**

Pinning down exactly which problem to solve, for whom, and where.

Read it three times today — it sticks!

1.2 Stakeholders: everyone the machine touches

A scoped problem names its **stakeholders** — every person or group who affects the system or is affected by it, whether they ever open the app or not. The Class 8 court taught you to find them; this year you **map** them, because their interests usually disagree, and the disagreements are exactly where ethical failures start. A stakeholder map puts each group in one of four rings: **decides** (controls the system), **operates** (runs it daily), **serves** (is supposed to benefit), and **bears** (carries the risk or the harm). Most failed AI systems share one symptom: the "bears" ring was empty on the day the plan was signed.

T9-02 Stakeholder Map

PAIR · 15 MIN

Scenario: the district plans an **exam-result predictor** that flags Class 9 students "at risk of failing" three months early, so schools can offer extra coaching. Map every stakeholder into the four rings — at least two per ring, and put at least one group that the planners might forget:

Ring	Stakeholder	What they want	What they risk
DECIDES			
OPERATES			
SERVES			
BEARS			
BEARS — the forgotten one			

The disagreement test: pick the two stakeholders whose wants clash hardest. State both sides in one line each:

WORD POWER · 2

Definition: **stakeholder**

Any person or group affected by a system, helpful or harmful.

Read it three times today — it sticks!

ANALYST'S DRILL · WHO IS IN WHICH RING?

Write D (decides), O (operates), S (serves) or B (bears) — some fit more than one ring; write both and say why:

For the crop-disease scanner run by a state agriculture office…	Ring(s)
the field officers who photograph suspicious leaves	
the smallholders whose crop the app judges	
the company that owns the model and its data	
the neighbouring farms whose fields were never photographed	
the agricultural university that validated the labels	

1.3 The system map: name every feature the machine sees

Between a scoped problem and a working model stands the question builders answer with a diagram: **what will the machine actually look at?** A **system map** lists the inputs — called **features** (you met features in Class 7) — that flow into the machine, the prediction that flows out, and the human decision that closes the loop. The skill is in the **naming**: "student information" is not a feature; "attendance percentage last term", "previous score", "homework submission rate" are. Vague features produce vague machines — and nobody can audit what nobody can name.

FEATURE OR NOT? — THE ANALYST'S TWO-SECOND TEST

A feature must be **measurable from records the system will actually see**. Mark each YES or NO, one reason per line:

Candidate feature	Feature? (Y/N)	Why
number of library books borrowed last term		
is generally hardworking		
minutes late to school, per month		
comes from a good family		

T9-03 System Map of Data Features

GROUP · 20 MIN

Return to the exam-result predictor. Draw its system map as a table: at least **five** features with their units, the prediction it outputs, and the human who must make the final call. Then attack your own map:

Feature (precise name + unit)	Where would this data come from?	Who might this feature unfairly miss or punish?

Prediction output:

The human who must decide before any action:

The third column is the audit. Which ONE feature would you delete, and why?

WORD POWER · 3

Definition: **system map**

A diagram naming every feature that flows in, the output, and the human decision.

Read it three times today — it sticks!

■ MYTH BUSTER

MYTH *"More features is always better."*

TRUTH Every extra feature feels like extra wisdom, but features bring baggage: a "parent's occupation" column may smuggle caste and class into a machine that will quietly repeat them; a "number of tuition hours" feature may punish every family that cannot afford them. Each feature must EARN its place twice — first by genuinely predicting the outcome, then by surviving the fairness audit ("who does this column punish?"). Five honest features beat forty borrowed ones. Analysts cut; hoarders collect.

1.4 Rule vs learn: the showdown

Before the mathematics, settle the deepest question in the field — **when do you even need machine learning?** Some problems are fully solved by rules a person can write: "if temperature above 38°, flag fever" needs no examples. Others — recognising a face, judging an essay, predicting a fare through traffic — drown the rule-writer in exceptions. The showdown mission makes you fight one problem both ways, so you feel the exact point where rules run out and learning begins.

WHEN WOULD RULES LOSE? — 60 SECONDS OF PREDICTION

Name one task in your school that rules could NEVER finish — and say in one line what the examples would have to teach instead:

Two machines on my street: one that should stay rule-based forever, and one that should learn — with one line of defence each:

THE EXCEPTION COUNT — HOW RULES LOSE, MEASURED

Take the school-fee-reminder rule:
"If a fee is unpaid after the 10th, send the reminder SMS."
Exceptions it would hit this month (scholarships, split payments, portal outages…):

Now the analyst's questions:
Who fixes the rule when an exception hurts someone?

At roughly how many exceptions a month would you re-build it from examples?

That number — the exception rate you are willing to maintain by hand — is the real border between rule-based and learning-based. Keep it for the showdown.

T9-04 Rule vs Learn Showdown

HANDS-ON LAB · PAIR · 15 MIN

The task: **decide whether a message to the class group is "urgent" or "not urgent".** Fight it both ways.

ROUND 1 — WRITE THE RULES

Write three if-then rules that decide urgent / not urgent. Be precise enough that a stranger could apply them:

The message that breaks my best rule:

ROUND 2 — LEARN FROM EXAMPLES

Now collect 10 labelled messages (real or invented): 5 urgent, 5 not. List only their FEATURES (length, punctuation, capitals, sender, time sent) — and say what pattern your 10 examples teach:

The case that still feels unfair either way:

The verdict: for THIS task, which approach would you ship — and what would have to be true about the world for the other one to win?

WORD POWER · 4

Definition: **rule-based system**

A program that follows if-then rules written entirely by people.

Read it three times today — it sticks!

WORD POWER · 5

Definition: **learning-based system**

A program that finds its own pattern inside examples.

Read it three times today — it sticks!

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can place AI, ML and DL inside each other — and place real products in the right box ○○○
- [ ] I can run a 4Ws canvas and produce a one-sentence problem statement ○○○
- [ ] I can map stakeholders into decides / operates / serves / bears — and find the forgotten ring ○○○
- [ ] I can name precise features for a system map and say who each feature might punish ○○○
- [ ] I can fight a task both by rules and by examples, and defend which one I would ship ○○○

■ THINK & INK

A system I know that runs on rules nobody questions. If we rebuilt it as a learning system, the first thing that would change is …, and the risk would be …

CASE NOTES — WHAT CHANGED IN MY THINKING?

Before this chapter I believed all AI was one kind of clever. Now I can name the great divide — …

CHAPTER 1 CLUES — POCKET THESE

- **AI ⊃ ML ⊃ DL** — nested boxes; the great divide is **rule-based vs learning-based**, and it cuts across all three.
- **Problem scoping** is the 4Ws canvas — solutions discussed before scoping are guesses in disguise.
- Stakeholder rings: **decides, operates, serves, bears** — an empty bears-ring is how harm gets planned.
- A **system map** names precise features — and every feature must survive the fairness audit.

The clue I would tell my family tonight:

Chapter 1 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

1. A voice assistant and a lift are both "AI" in the widest sense. Name the ONE question that separates them — and give each machine's answer.

2. In the exam-predictor case, which ring did the "forgotten" stakeholder belong in, and what specifically did they risk?

3. Why is "student information" not a feature — and what makes "attendance percentage last term" one?

MY CASE JOURNAL — TODAY'S SHARPEST CLUE ABOUT SCOPING AND STAKEHOLDERS

2. Data Literacy

GUIDING QUESTION

Can I trust this number — and can I trust the picture it comes in?

Learning systems eat data; this chapter teaches you to be the food inspector. You will sort data by type and structure, audit a source for quality and motive, catch the three classic ways graphs lie — truncated axes, cherry-picked windows, and pies that add to more than 100% — and separate privacy from security, two ideas that are constantly confused. The chapter ends with a dashboard you build yourself, where every chart must survive your own cross-examination.

MISSIONS T9-05 – T9-09
OUTCOMES 9.D1 · 9.D2 · 9.R1 **STRANDS** D R

Links: Maths — statistics & graphs · Social Science — sources & media · English — critical reading

In this chapter

2.1 Know your data: type and structure 21
2.2 Source quality: who says so, and why do they care? 22
2.3 The misleading graph gallery: three classic tricks 23
2.4 Privacy vs security: the constant confusion 25
2.5 The paper dashboard: three charts, one story 27

TERMS YOU'LL ANALYSE WITH

qualitative data · quantitative data · unstructured data · privacy · security

WARM-UP · BEFORE YOU READ ALONE · 2 MIN

A news app shows: "Exams easier this year — 68% agree". Write the FIRST question you would ask before believing it. Compare with your answer on the last page of this book.

ANALYST'S TOOLKIT — YOU WILL NEED: pencil · ruler · a die and a coin · your sharpest suspicion

2.1 Know your data: type and structure

Before any judgement, classify what you are holding. **Qualitative data** describes qualities — names, categories, descriptions ("mango", "delayed", "angry"). **Quantitative data** counts or measures — numbers you can average, plot, and compare ("42 marks", "12 minutes"). Then a second cut: **structured data** sits in neat rows and columns a machine can read directly (your attendance register); **unstructured data** does not — photographs, voice notes, essays, chat messages — and must be converted into features before any machine can learn from it. Most of the data in the world is unstructured, which is exactly why feature-naming (Chapter 1) is a superpower.

T9-05 Data Type Sort

■ QUICK WIN ALONE · 10 MIN

Sort each item twice: QL (qualitative) or QT (quantitative) — and S (structured) or U (unstructured):

The item	QL or QT?	S or U?	If U — one feature that would make it learnable
a voice note complaining about the canteen			
the class attendance register for March			
"the bus was very late today" (written feedback slips)			
blood-sugar readings from a clinic, one per patient per day			
500 photos of street intersections			
marks out of 100 for the unit test			

The pattern I notice about the U column:

WORD POWER · 6

Definition: **qualitative data**

Data describing qualities or categories, like names and descriptions.

Read it three times today — it sticks!

WORD POWER · 7

Definition: quantitative data

Data measured in numbers, ready to count, average and plot.

Read it three times today — it sticks!

WORD POWER · 8

Definition: unstructured data

Data not in rows and columns — photos, audio, free text.

Read it three times today — it sticks!

2.2 Source quality: who says so, and why do they care?

A number is only as trustworthy as its **source** — and a source has three faces you must check: **competence** (do they actually measure this, or just repeat it?), **method** (how was it counted — and how many, over how long, leaving out whom?), and **motive** (what do they gain if you believe it?). None of the three checks requires a computer; all three require healthy suspicion. A government census and a phone-company advert may quote the same "average income" with completely different honesty.

SOURCE SELFIE — RATE ONE SOURCE YOU ACTUALLY MET THIS WEEK

The source: _____ — competence 1–5: _____ method 1–5: _____ motive 1–5: _____

The one line that decided my lowest score:

Motive detector, one glance: this source WINS if I believe it because …

TODAY'S NUMBER, THREE FACES

Copy one number-claim from this week's news and run the three-face check on it — competence, method, motive — then name the face that decided you:

claim _____ h(4pt)
faces competence _____ method _____ motive _____ h(3pt)
decider _____ h(4pt)

T9-06 Source Quality Audit

PAIR · 15 MIN

Your school's notice board shows three claims about the same thing — students' travel time to school. Audit each source on the three faces (rate each 1–5, with a one-line reason), then give your verdict:

THE THREE CLAIMS

Claim A (school notice board): "Average travel time is 34 minutes" — measured by Class 9-B from a survey of 40 students, one morning, during exam week. **Claim B (a bus company's flyer):** "Most students reach school in under 20 minutes by bus!" — no method stated, next to the price list. **Claim C (the district education report):** "Median travel time across 12 schools: 41 minutes" — full method in appendix, surveyed across a full term.

Face of the source	Claim A (Class 9-B)	Claim B (bus company)	Claim C (district report)
Competence 1–5			
Method 1–5			
Motive 1–5			

The verdict — which number would you quote in a class debate, and in what words?

Claim A is honest work but weak method. Write the ONE extra line its collectors should have added:

2.3 The misleading graph gallery: three classic tricks

Charts feel like evidence, but a chart is an **argument drawn by someone**. Three tricks do most of the lying. **Truncated axes**: the bar chart's y-axis starts at 88, so a 2% difference looks like a cliff. **Cherry-picked windows**: the line chart shows exactly the weeks that prove the point, hiding the weeks that don't. **Pie arithmetic**: the "shares" add to more than 100%, which is impossible — or the categories overlap. None of these tricks requires a dishonest machine; the machine will happily draw whatever it is told. Your defence is the same three questions every time: **where does the axis start? what is NOT shown? do the parts add up?**

T9-07 Misleading Graph Gallery

GROUP · 20 MIN

The gallery shows two charts from the same school canteen data. Chart A (honest) and Chart B (same data, "improved" by the vendor's design team). Study them, then complete the charge sheet.

CHART A — AS THE CLERK DREW IT

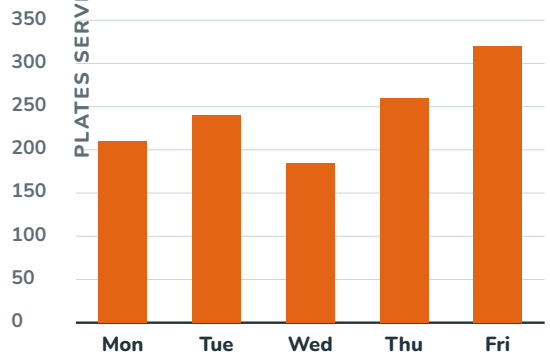
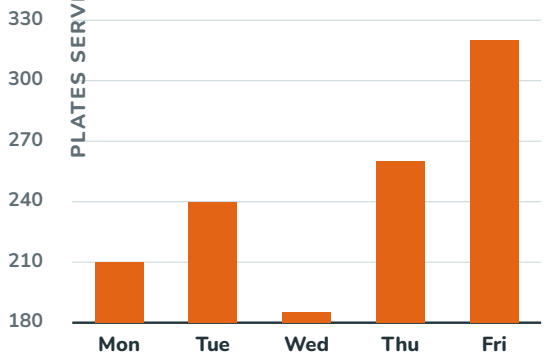


CHART B — AS THE VENDOR DREW IT



The charge	Chart A or B?	The exact evidence (read the axis!)
The y-axis starts above zero, exaggerating every change		
Tuesday looks like a disaster; on honest axes it is a mild dip		
Same data, completely different story		

Second exhibit — the pie that broke arithmetic: a "share of canteen spending" pie shows snacks 45%, drinks 35%, lunch 55%. **What is impossible here, and what did the drawer probably do wrong?**

Your three-question defence, in order:

ANALYST'S DRILL · NAME THE TRICK

Each description hides one of the three classic tricks — or none. Write T1 (truncated axis), T2 (cherry-picked window), T3 (broken pie arithmetic) or OK:

The chart	Trick?
A rainfall chart whose axis starts at 200mm, so a "drought" year looks like none	
An attendance line that shows only exam weeks	
A "how students spend their day" pie: sleep 40%, school 30%, study 20%, screens 25%	
A bar chart of marks with the axis starting at zero and equal steps	
A price-rise chart comparing only the two months before an election	

2.4 Privacy vs security: the constant confusion

Two words, one wall between them, and most people — including many adults — use them as if they were the same. **Privacy** is your **control**: who may know what about you, and on what terms. **Security** is the **protection**: the locks, backups and rules that keep data away from people who should never have it. You can have security without privacy — a perfectly guarded file nobody consented to — and privacy without security — a promise nobody enforced. Learning systems touch both: they need **features** (which may be personal), and they run on **infrastructure** (which may leak).

SORT FIRST — P OR S?

The situation	P or S?	The wall between them
A hospital's records are encrypted but shared with an insurer without asking		
A family photo album sits on an unlocked family laptop		
A learning app stores marks safely — and reads chat logs it never needed		

The habit: security is judged by LOCKS, privacy by PERMISSIONS — a system needs both, separately audited.

T9-08 Privacy vs Security Scenarios

PAIR · 15 MIN

For each scenario: does it fail PRIVACY (P), fail SECURITY (S), or both (B)? Then write the one fix:

The scenario	P / S / B?	The one fix
The school's mark database sits on an unlocked, shared laptop		
The canteen app quietly sells order histories to a snack company		
A "free" photo-print kiosk keeps every photo on a drive anyone can open		
The exam cell emails full mark sheets, names visible, to the wrong list		
A learning app reads students' locations at all hours for "service quality"		

The two-sentence rule I will use from now on — privacy is …, security is …

WORD POWER · 9

Definition: **privacy**

Your control over who may know what about you, and on what terms.

Read it three times today — it sticks!

WORD POWER · 10

Definition: **security**

The locks and rules that keep data away from people who should not have it.

Read it three times today — it sticks!

2.5 The paper dashboard: three charts, one story

Analysts communicate with **dashboards**: small groups of charts chosen so that together they tell one honest story. The craft is in the choices — which chart for which job (bars to compare, lines to show change over time, pies only for parts of a whole that truly add to 100%), which time window, which axis start. You will build one on paper about your own class's week, and then do the step most dashboards skip: **write the story the charts tell, including the part they cannot show**.

CHART-CHOICE WARM-UP — RIGHT CHART FOR RIGHT JOB

The message to show	Bar / Line / Pie?	The one-line reason
"How the five houses' collection totals compare"		
"How the temperature moved across the fortnight"		
"What share of waste is food, plastic, paper"		
"Attendance on each day of the week"		

A pie is the most abused chart in the world: it may only ever show parts of ONE whole that sum to 100%. When in doubt, choose the bar.

AXIS START SCAN — THE FIRST GLANCE AT ANY BAR CHART

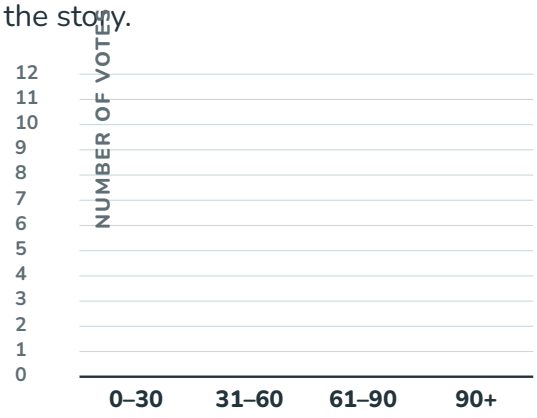
The chart says…	Where does its y-axis start?	Honest or truncated?
"Marks jumped this year!" — bars look like a cliff		
"Attendance dipped slightly" — bars look almost equal		

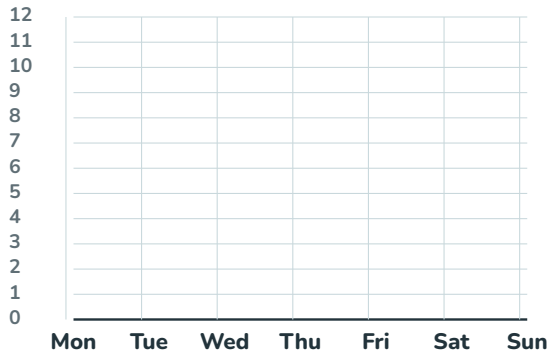
Rule of thumb: bar charts start at zero; line charts may zoom — but must SAY so on the axis.

T9-09 Paper Dashboard

HANDS-ON LAB GROUP · 25 MIN

Your team surveys the class on three questions: **minutes of screen time yesterday**, **marks on the last unit test**, and **mode of travel to school**. Collect quick tallies (your teacher will run the vote), then draw all three charts below — right chart for right job — and finish with the story.





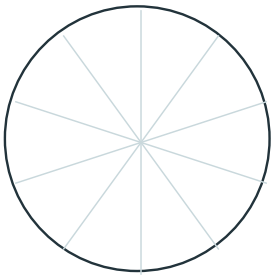


Chart 1 (bar): screen minutes — I chose a bar chart because …

Chart 2 (line): marks across the week's revisions? State what the line shows:

Chart 3 (pie): travel modes — my shares add up to exactly … %

The story in three sentences — including one thing all three charts CANNOT show:

MYTH BUSTER

MYTH *"Average = typical."*

TRUTH The average is one summary number, and every summary throws something away. In the town where the district report claimed a 41-minute median travel time, one village bus ride of 150 minutes sits next to a five-minute walk — the **mean** would say "about 45 minutes", which describes nobody. When the distribution is lopsided — incomes, travel times, hospital waits — the **median** (the middle person) tells you about the typical person and the **mean** tells you about the total. Neither is "the truth"; each answers a different question. Analysts name which one they used — and so will you, from the next chapter onward.

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can sort any dataset by type (QL/QT) and structure (S/U) — and name features for the U column ○○○
- [ ] I can audit a source on competence, method and motive — and defend my ratings ○○○
- [ ] I can catch truncated axes, cherry-picked windows and broken pie arithmetic on sight ○○○
- [ ] I can explain privacy and security as two different failures — with one fix for each ○○○
- [ ] I can build a three-chart dashboard and write the story it tells, including its blind spot ○○○

THINK & INK

A chart or claim I believed before this chapter. The trick it used (or the check it failed) was …

HOME LINK *take this page home — no screen needed*

AT HOME Headline Autopsy

HOME · 15 MIN

With an adult, take ONE number from this week's news (TV, paper, or a forwarded message — do not open links). Run the source audit: competence, method, motive. Then ask the three chart questions if a picture came with it.

The number and its headline:

The three faces checked (C / M / M), with one line each:

The one question we would send to whoever published it:

CASE NOTES — WHAT CHANGED IN MY THINKING?

Before this chapter I believed a chart or a number was neutral. Now my first three questions are …

CHAPTER 2 CLUES — POCKET THESE

- Classify first: **QL/QT**, then **S/U** — unstructured data needs features before it can teach.
- Audit every source three ways: **competence, method, motive** — motive never sleeps.
- Graph tricks: **truncated axis, cherry-picked window, broken pie arithmetic** — three questions catch all three.
- **Privacy** is control, **security** is protection — a system can fail either one alone.

The clue I would tell my family tonight:

MY WEEK OF NUMBERS — where I will audit first: One number I will re-examine this week, the source it came from, and the face I now doubt most:

Chapter 2 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

1. Why can a survey of 40 students in exam week be honest work with weak method? Name the missing line.

2. A pie chart's shares are snacks 45%, drinks 35%, lunch 55%. What is impossible — and what probably went wrong?

3. "The database is perfectly guarded, so privacy is fine." Where exactly does this argument break?

MY CASE JOURNAL — TODAY'S SHARPEST CLUE ABOUT DATA THAT CAN AND CANNOT BE TRUSTED

3. The Maths Behind AI (on paper)

GUIDING QUESTION

What does the machine actually compute — and when does the computation fool it?

This is the engine room. Five missions, five mechanisms, all by hand: the three averages and the spread they hide (mean, median, mode, range), probability you can roll on your desk, a line of best fit you draw through real dots and then interrogate, nearest-neighbour classification with distances as evidence, and the pattern-to-rule move that turns a table into an algorithm. Every machine-learning system in the world is some arrangement of exactly these ideas — run by silicon instead of by you. By the end of this chapter, "the AI predicted" will mean something you could check.

MISSIONS T9-10 – T9-14
OUTCOMES 9.M1 · 9.M2 · 9.M3 · 9.M4 · 9.T1 **STRANDS** L

Links: Maths — statistics, probability, coordinate geometry · Science — experiment & error · English — precise claims

In this chapter

3.1 Three averages and a liar 33
3.2 Probability: the mathematics of 'how sure' 35
3.3 The line of best fit: drawing the trend, then doubting it 37
3.4 Nearest neighbours: classification with a ruler 39
3.5 Pattern to rule: write the algorithm 41

TERMS YOU'LL ANALYSE WITH

mean · median · mode · probability · correlation · line of best fit · nearest neighbour

WARM-UP · BEFORE YOU READ ALONE · 2 MIN

Five students wait 5, 6, 7, 8 and 90 minutes for the bus. Without calculating, is the "average wait" nearest to 6 or to 23? Why did your gut hesitate?

ANALYST'S TOOLKIT — YOU WILL NEED: pencil · ruler · a die · a coin · graph-squared paper if you have it

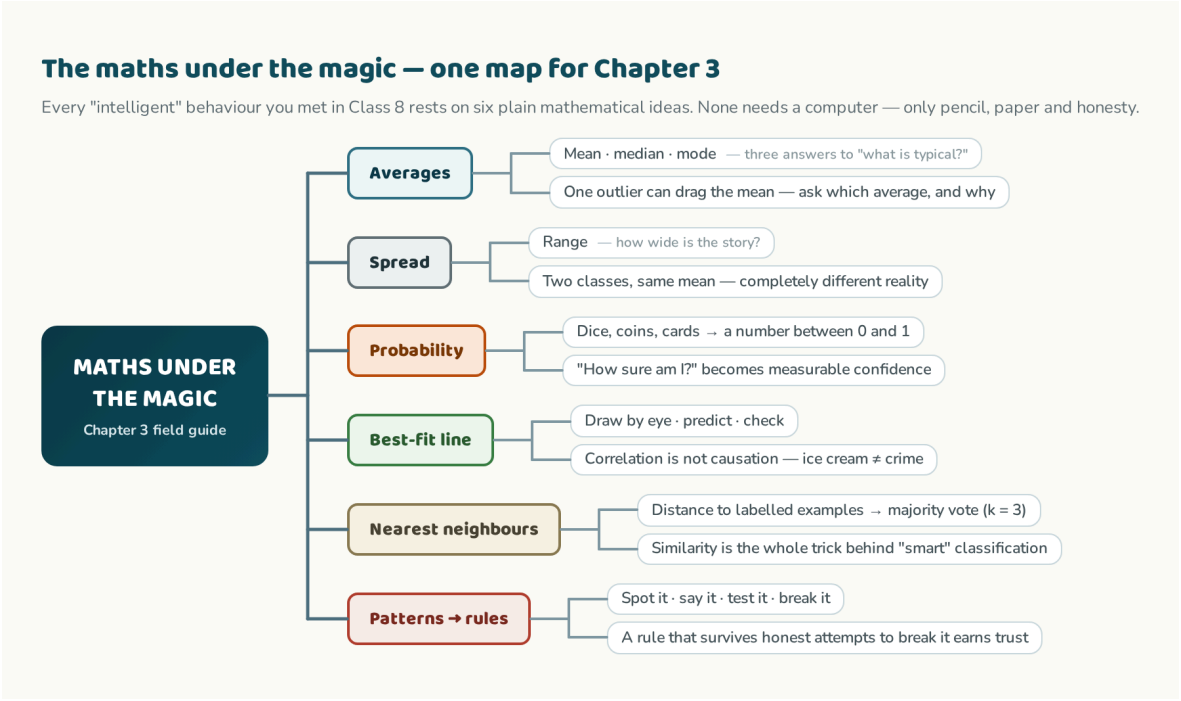


FIG Chapter 3 as one map: six plain ideas, each met through an experience before a definition.

3.1 Three averages and a liar

You met averages long ago; now meet their arguments. The **mean** adds everything and divides — sensitive to every value, especially extremes. The **median** stands in the middle of the sorted list — blind to how extreme the extremes are. The **mode** is the most frequent value — the only average that works on categories. And the **range** (highest minus lowest) is the crudest measure of **spread** — how scattered the data is. A machine summarising “its users” faces exactly these choices, and each choice can be dressed up as THE answer. The analyst’s defence is to ask: **which average, and what does it throw away?**

THREE AVERAGES, THREE JOBS — PRE-COMPUTE, THEN ARGUE

Average	Best job	Its signature failure
Mean	fair shares and totals — “per person, per week”	one outlier drags it across the story
Median	typical case in a skewed crowd — incomes, wait times	ignores how big the top really is
Mode	what actually happens most — sizes, routes, defaults	useless when everything is nearly unique

Before T9-10's full example: pick which average you would bring to a housing-society rent row, and say why in one line — the table above is the hint.

T9-10 Average Lies

■ QUICK WIN ALONE · 15 MIN

A startup advertises: “Our employees earn ₹62,000 a month on average!” The nine salaries (in ₹1000s): 22, 24, 25, 26, 28, 30, 32, 35, 268.

The average	Your calculation (show the working)	What it really describes
Mean		
Median		
Mode		
Range	—	

Which average is the advertisement using — and which one should a job-seeker hear?

The founder says: “but it IS the average!” Write the one-sentence reply that is true, fair and devastating:

Delete the outlier (268) and recompute the mean in the margin. What does this tell you about one extreme value’s power?

WORD POWER · 11

Definition: **mean**

The sum of all values divided by how many there are.

Read it three times today — it sticks!

WORD POWER · 12

Definition: **median**

The middle value when the data is sorted in order.

Read it three times today — it sticks!

WORD POWER · 13

Definition: mode

The value that appears most often in the data.

Read it three times today — it sticks!

ANALYST'S DRILL · WHICH AVERAGE WOULD YOU USE?

Pick the average that answers the question most honestly — mean, median or mode — and defend it in a line:

The question	Mean / Median / Mode?	The defence
"Which shoe size should the school shop stock the most of?"		
"What is a typical app-learning session length, when a few marathons of 6 hours exist?"		
"Split the class into three fair teams by marks — what should each team's centre be?"		
"How different are the fastest and slowest bus runs?"		

*Notice the fourth row: range is not an average at all — it is **spread**. Knowing which question needs which tool is the analyst's first instinct.*

3.2 Probability: the mathematics of 'how sure'

A machine that says "92% confident" is making a **probability** claim — and probability is something you can hold in your hand. The basic deal: probability = **favourable outcomes ÷ all equally likely outcomes**, so a fair die shows a six with probability 1/6. But the deeper idea is **evidence**: if you roll a die 60 times, you EXPECT about 10 sixes — and you will rarely get exactly 10. Probability describes the long run, not the next throw. That gap between "expected" and "observed" is where a healthy analyst lives: too-perfect streaks are suspicious, wild streaks are normal, and **confidence** without a count behind it is decoration.

T9-11 Likelihood Lab

■ HANDS-ON LAB PAIR · 20 MIN

Predict first, then test. No calculator — just tallies.

Part A — the die. My prediction for 60 rolls: number of sixes = … , number of odds = … Now roll 60 times (or split the work in your pair and combine). Tally:

Outcome	Tally	Count	My expectation	How far off?
6			10	
1, 3, 5 (odds combined)			30	

Part B — the coin and the sneaky question. A coin lands heads 4 times in a row. My next flip is… (tick one) ☐ more likely tails ☐ more likely heads ☐ exactly 1/2 either way **The reason, in one sentence:**

Part C — the machine's confidence. A classifier says "92% sure this leaf is diseased." Using your dice experience: if we tested it on 100 similar leaves, how many would you EXPECT it to get right — and would you be shocked at 84?

WORD POWER · 14

Definition: **probability**

Favourable outcomes divided by all equally likely outcomes.

Read it three times today — it sticks!

ANALYST'S DRILL · WHAT IS THE REAL PROBABILITY?

The setup	Favourable ÷ all	As a percentage
A bag has 3 red, 4 blue, 5 green beads. Red?		
Two coins flipped together. Both heads?		
A standard die shows a number 4 or more?		
A "90% accurate" screening machine flags a healthy person. Chance of that on any one test — if 20 of 100 people are actually unhealthy?		

The last row is the honest one: even a good machine misfires sometimes, and knowing the base rates — who is actually in the pile — is what stops a percentage from scaring you.

3.3 The line of best fit: drawing the trend, then doubting it

Here is regression at analyst level. Scatter real data — say, hours of study against marks — and a cloud of dots appears. Draw the one straight line that hugs the cloud as fairly as possible: not chasing the extremes, ignoring no dot, roughly equal numbers of dots above and below. That line is a **model**: it predicts a mark for any study time, including times you never observed. But two warnings come free with it. **Extrapolation** — predicting far outside the dots — is guesswork wearing a suit. And **correlation** — the pattern itself — never proves **causation**: study hours may rise with marks because quieter students both study more and test better. The line shows that two things move together; only an experiment can say who moves whom.

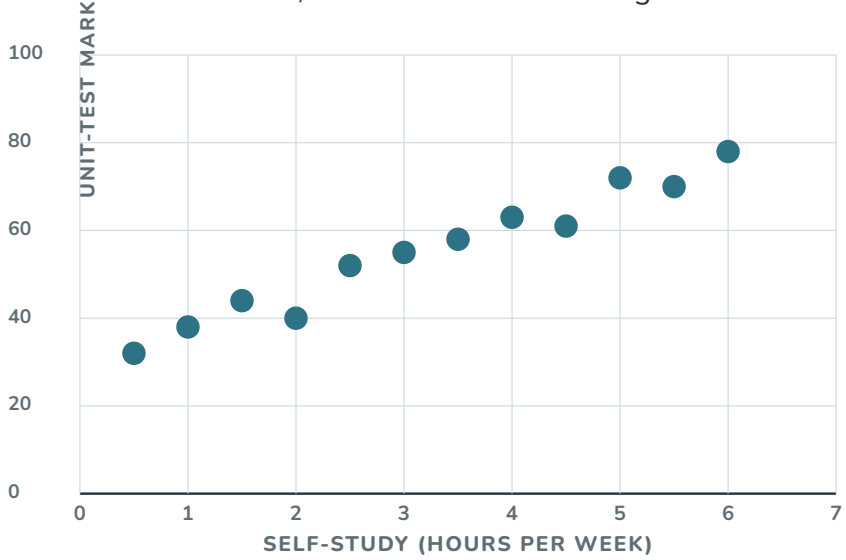
EXTRAPOLATION SENSOR — CALIBRATE IT BEFORE YOU DRAW

A line fitted on 0–6 study hours is asked about a student who studies 12 hours. In one line: what is the HONESTEST thing your line can say about that student?

T9-12 Line of Best Fit by Eye

HANDS-ON LAB · GROUP · 20 MIN

The scatter shows twelve students: hours of self-study per week (x) against unit-test marks (y). Draw the best-fit line with a ruler, then answer the interrogations.



Prediction 1 (inside the dots): a student who studies 3.5 hours — my line predicts about … marks. **Prediction 2 (extrapolation — be brave):** 10 hours a week predicts about … marks. **Why is this prediction the weakest of the two?**

The causation trap: "more study CAUSES higher marks" — give one alternative explanation that fits the same dots:

Another team's line differs from yours by 4 marks. Whose is right? What does this tell you about drawing lines by eye — and about machines that must choose ONE line?

Definition: line of best fit

The straight line that follows a data cloud most fairly, used to predict.

Read it three times today — it sticks!

WORD POWER · 16

Definition: correlation

Two measures moving together — which does not prove one causes the other.

Read it three times today — it sticks!

ANALYST'S DRILL · CORRELATION OR CAUSATION?

Each claim spots a real correlation — then smuggles in causation. Mark C (causation claimed but unproven) or OK, and name the sneaky alternative where you can:

The claim	C or OK?	The sneaky alternative
"Villages with more AI-irrigation have higher yields — irrigation works!"		
"Umbrella sales rise with rainfall."		—
"Classes with more projectors score higher — projectors teach!"		
"Ice-cream sales and drownings rise together — ban ice-cream!"		

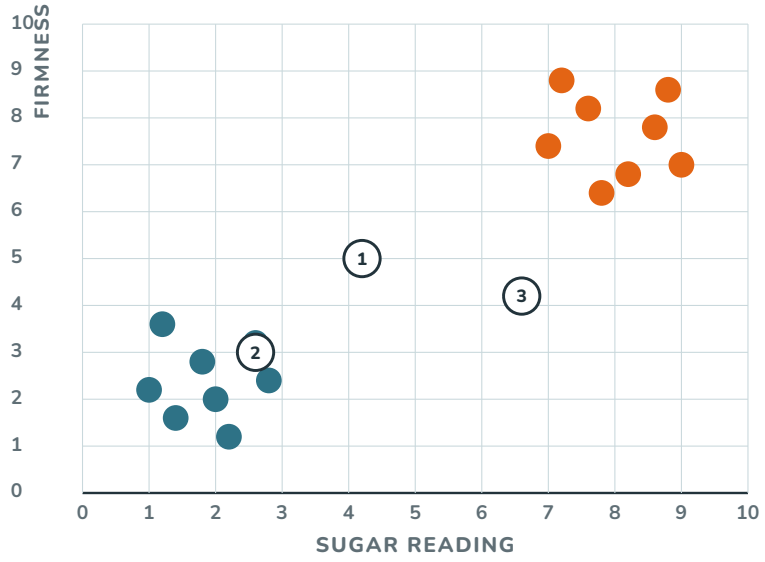
3.4 Nearest neighbours: classification with a ruler

You ran a nearest-example classifier in Class 8; now give it its adult name and its arithmetic. To classify a new point, a **k-nearest-neighbour** system measures the distance from the new point to labelled examples, keeps the **k** closest, and lets them vote. Distance on a grid has an exact recipe — the square root of (x-gap squared + y-gap squared) — but the analyst's ruler version (count the squares, compare) captures every idea except the square root. The two honest weaknesses: **scale** (a feature measured in hundreds bullies a feature measured in ones) and **k** (one neighbour is a gossip; a large k is a crowd — the vote changes with k).

T9-13 Neighbour Vote

HANDS-ON LAB · PAIR · 25 MIN

The plot shows the mango-sorter's training data: teal dots = ripe (sugar reading × firmness), amber = not yet. Three new mangoes arrive. Use a ruler: find each new point's THREE nearest labelled dots (that is k = 3), note their labels, and let the majority vote.



New mango	My 3 nearest neighbours (labels)	The vote	Final classification
1 (4.2, 5.0)			
2 (2.6, 3.0)			
3 (6.6, 4.2)			

Mango 3 is the border case — its nearest neighbours are far away, and the vote is close. Which k would you ship to the depot, and how would you tell the depot manager what the machine still gets unsure about?

The scale trap: suppose sugar readings went up to 400 instead of 10. Would the ruler still find the same neighbours? What should the builder do to both features before measuring?

WORD POWER · 17

Definition: nearest neighbour

Classifying a new point by the vote of its k most similar labelled examples.

Read it three times today — it sticks!

MYTH BUSTER

MYTH *"The machine found the pattern, so the pattern is real."*

TRUTH A learning system finds **a** pattern — the one its features and its data support best, with the k and the line it was told to use. Change the features, the sample or the k, and a different "pattern" appears, honestly computed all the way. The mango sorter with sugar-and-firmness finds one rule; with weight-and-days-since-picking it finds another; both are true to their data. Patterns are **manufactured from choices** — which is why this book keeps asking you to write your choices down. The computation is real; the pattern is a **decision**. Analysts sign their decisions.

3.5 Pattern to rule: write the algorithm

The final mechanism is the analyst's party trick: turn a data table into an **algorithm** — a decision procedure so precise that another person (or a machine) can run it without you. This is the bridge between statistics and code, and you will cross it with no code at all: just a rule, an ordered checklist, and the honesty to state what your rule gets wrong.

THE RULE-WRITER'S HABIT — READ IT BEFORE T9-14

A good rule has three marks: **ordered** (the checklist runs top to bottom, no jumping), **complete** (every case lands SOMEWHERE), and **confessed** (the writer names one case it treats badly). Keep all three in sight as you write yours:

Field study first: the lost-property box has 12 items in T9-14. Before reading them, predict which feature will separate the piles best — and which pair of items will be hardest to split:

T9-14 Pattern to Rule

ALONE · 15 MIN

The table shows the school lost-property box: 12 items, four features each. Study it, then write a decision rule that sorts any new item into KEEP-DESK / RETURN-CLASS / LOST-AND-FOUND.

Item	Valuable?	Named?	Found where?
water bottle	no	no	corridor
spectacles case	yes	no	classroom
geometry box	yes	yes	classroom
single slipper	no	no	playground
library book	no	yes	canteen
earphones	yes	no	canteen
lunch box	no	yes	corridor
school ID card	no	yes	playground

My decision rule (if-then, in order, ready for a stranger to run):

Now trace it: a costly unnamed watch is found on the playground. My rule sends it to … , because …

The case my rule handles unfairly (write it yourself — every rule has one):

AT A GLANCE · YOUR PAPER MECHANISMS VS THE MACHINE'S

The mechanism	You did it with…	The machine uses…	What never changes
Mean, median, mode, spread	pencil division	the same arithmetic, at scale	every summary throws something away
Probability	dice and tallies	counts at enormous scale	long-run expectations, not promises
Best-fit line	ruler and eye	least-squares minimisation	extrapolation is still guesswork
k-nearest neighbours	ruler and vote	distance formulas + fast search	scale and k still decide the answer

*The right-hand column is the whole point: the **judgement** does not shrink when the computer grows. That is why analysts — not just machines — are hired.*

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can compute mean, median, mode and range — and say which one a claim is hiding behind ○○○
- [ ] I can predict a probability, test it with dice or coins, and reason about the gap ○○○
- [ ] I can draw a best-fit line, predict with it, and explain the extrapolation and causation traps ○○○
- [ ] I can run k-nearest-neighbour classification with distances, and defend my choice of k ○○○
- [ ] I can turn a data table into a rule another stranger can run — and name one case it treats unfairly ○○○

■ THINK & INK

The mechanism that surprised me most this chapter was …, because before computing it by hand I believed …

HOME LINK *take this page home — no screen needed*

AT HOME **Household Statistics** HOME · 15 MIN

With an adult, pick one repeated household number — monthly electricity units, travel minutes to work, or weekly vegetable spend (past bills are fine; no need to measure anything new). Find the mean, median and range together, and hunt for the outlier.

The numbers (at least 6 periods):

Mean, median, range (show one division):

Which average describes our household best — and which one would an advertiser quote?

CASE NOTES — WHAT CHANGED IN MY THINKING?

Before this chapter I believed "the AI predicted" was a kind of magic. Now I can name the paper mechanisms underneath: …

CHAPTER 3 CLUES — POCKET THESE

- Every average answers a different question — **mean is sensitive, median is stubborn, mode counts what repeats**.
- **Probability** describes the long run; a streak is not a promise, and confidence without a count is decoration.
- The best-fit line predicts inside the dots; **extrapolation is guesswork, correlation is not causation**.
- kNN votes by distance — **scale and k** decide the answer, so the analyst writes their choices down.

The clue I would tell my family tonight:

Chapter 3 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

1. Salaries: 22, 24, 25, 26, 28, 30, 32, 35, 268 (in ₹1000s). Compute the median — and explain why the 268 barely moves it.

2. A fair die is rolled 60 times. How many sixes do you EXPECT, and why is getting exactly that not guaranteed?

3. Study hours vs marks: the line predicts 10 hours → 95 marks. Name BOTH traps in that prediction.

4. In T9-13, mango 3 changed its vote when k changed from 1 to 3. What does that teach about "the machine's answer"?

HUMAN VS MACHINE **Pattern vs proof** HUMAN AI

The nearest-neighbour vote finds an answer; a proof explains why the answer must be true. This chapter you did both by hand — now name the difference.

THE HUMAN WAY

Reasoning carries beyond the data: prove the angle sum of a triangle once and it holds for every triangle that ever will exist — even ones nobody will ever measure.

THE MACHINE WAY

The k-vote is statistics: only as good as its examples, silent about its "why", and confidently wrong outside its data with no way to notice.

One question from this chapter where your REASONING beats the pattern-vote — and why:

A machine with a million labelled triangles still cannot prove the angle sum. What exactly is missing?

Where would you still prefer the machine's vote over a human hunch? Defend it with a number:

MY CASE JOURNAL — TODAY'S SHARPEST CLUE ABOUT THE MATHEMATICS UNDER THE MACHINE

4. Generative AI & Ethics

GUIDING QUESTION

Machines that write, speak and show faces — how do they work, and what do we owe each other?

The strangest machines in the AI city make new things: essays, images, voices, faces. This chapter builds a text generator's core trick on your own desk — a next-word table you construct and then "generate" from — so the mystery becomes a mechanism. Then the ethics floor: real-or-AI evidence checks, deepfakes, copyright and academic integrity, the access divide, and a balloon debate where AI's future must argue for its own seat on the lifeboat. You leave with a verification routine you can run on anything any machine ever hands you.

MISSIONS T9-15 – T9-18
OUTCOMES 9.W1 · 9.R1 **STRANDS** W R

Links: English — language patterns & argument · Social Science — access & equity · Art — what is authorship?

In this chapter

4.1 Generative AI: the author that predicts 48
4.2 Build the next-word machine — then watch it invent 48
4.3 Deepfakes and the verification routine 50
4.4 The Verify-It routine: four moves, no excuses 51

TERMS YOU'LL ANALYSE WITH

generative AI · deepfake · verification routine · bias

WARM-UP · BEFORE YOU READ ALONE · 2 MIN

"Generative AI looks things up in a database of facts." — True or false? Write your verdict and ONE reason. You will re-judge this after Task T9-16.

ANALYST'S TOOLKIT — YOU WILL NEED: pencil · the printed sample texts in T9-15 · sticky notes for the debate · a fair-minded heart

4.1 Generative AI: the author that predicts

Generative AI means systems that produce new content — text, images, audio, video — rather than only sorting or predicting numbers. The idea is easiest to see in text. A text generator reads the words so far and predicts a **likely next word**, one step at a time, each prediction shaped by patterns in a mountain of human writing. Crucially — and this is the misconception this book exists to kill — it does not **look facts up**. There is no archive of truth inside it. There are patterns: which words tend to follow which, in which tone, for which audience. When those patterns line up with facts, the output reads like an encyclopedia. When they don't, it reads **exactly as confident** — about something that never happened. You are about to build the smallest possible version of this machine, and watch it invent.

T9-15 Real or AI?

■ QUICK WIN CLASS · 15 MIN

Your teacher shows four texts about the same topic (printed, or read aloud). Some are human-written, some machine-generated. Before any reveal, run the evidence checklist on each — not "does it sound smart?" but the three checks below:

Check	What to look for	Text 1	Text 2	Text 3	Text 4
Specificity	Real names, dates, local details you could verify — or smooth generalities?				
Sensory grounding	Does it mention what only a person present would know?				
Error texture	Human text has uneven emphasis and small tangents; generated text glides				
My verdict — human or AI?	commit BEFORE the reveal				

After the reveal: the check I trusted most was …, and the case that fooled me teaches …

The honest conclusion about "sounding real":

4.2 Build the next-word machine — then watch it invent

Time to be the machine. A next-word table is the smallest honest model of a text generator: for each phrase-ending, count which word followed it in some body of text, and the counts become

the probabilities. You will build one from a short corpus, then generate sentences from it. The inventions you will see are not bugs in your table — they are the mechanism working exactly as designed.

T9-16 How a Text Generator Guesses

■ HANDS-ON LAB PAIR · 20 MIN

Step 1 — the corpus. Here is your training text (a fictional village news column):

> *"The school team won the final match. The village celebrated the final match with sweets. The monsoon filled the lake. The team captain thanked the village. The monsoon delayed the final match. The captain smiled. The village repaired the school roof. The school thanked the village."*

Step 2 — build the table. Count what follows each phrase-ending in the corpus:

After the words…	Count of each next word	The winner (and its share)
"The school"		
"The monsoon"		
"The team"		
"The captain"		
"The village"		

Step 3 — generate. Start with "The school" and let the table choose the next word (highest count wins; on ties, choose either). Write your generated sentence:

Step 4 — the verdict. Is your generated sentence (a) always true to the village, (b) sometimes strange, or (c) both? Where did strangeness enter — and would a BIGGER table fix it completely?

Revisit your warm-up verdict: "generative AI looks things up" — now what do you say?

Definition: generative AI

Systems that create new text, images, audio or video from patterns.

Read it three times today — it sticks!

ANALYST'S DRILL · THE THREE BUILT-IN LIMITS

Match each symptom to its mechanism — H (hallucination: patterns without truth), B (bias: patterns of the past, including unfair ones), or C (copyright: patterns taken from people's work without permission or pay):

The symptom	H / B / C?
A "biography" of a teacher who does not exist — confident, detailed, false	
A recruitment tool trained on ten years of hires repeats their one-sidedness	
An image generator draws a character suspiciously like a famous comic artist's style, unpaid	
A chatbot cites a court case that no court ever heard	
A voice model clones a singer's voice for an advert she never approved	

Notice: none of these is a "bug" that a smarter machine removes. Each is the pattern-matching mechanism meeting the world it was fed.

4.3 Deepfakes and the verification routine

When generation meets a person's face or voice, it becomes a **deepfake** — synthetic media that looks or sounds like a real person saying, doing, or wearing things they never did. The technology is improving faster than most people's defences, which makes your defence a **routine**, not a feeling. The Class 8 three moves — trace, test, smell — grow a fourth here: **seek the original** (can you find the unedited source or the person's own account?). And remember the asymmetry: a deepfake needs only to exist for hours to do harm; your verification needs minutes. Run it anyway — minutes are how trust survives.

DEEPFAKE DEFENCE — THE TWO QUESTIONS THAT END MOST OF THEM

Ask…	Why it works
"Where is the FIRST place this appeared — and who posted it there?"	Fakes travel by re-upload; their first home is rarely traceable
"Does the real person's OWN account (or office) confirm it?"	The person's own voice is the one recording no fake can out-shout

A deepfake must only survive long enough to be forwarded. Your two questions take two minutes — that is the whole arms race.

T9-17 Balloon Debate

GROUP · 25 MIN

The balloon is sinking; only one speaker can stay aboard. Four speakers each defend a position on **"AI in our school"** — but each must argue from a DIFFERENT stakeholder's view, and the audience scores the **reasoning**, not the popularity:

Speaker (role)	The position to defend	Strongest point made	Weakest point exposed
A — a student who uses a chatbot for first drafts	"Generative AI is my tutoring partner."		
B — a teacher who grades the writing	"It threatens how we learn to write."		
C — a parent paying for internet data	"Access is unequal; that decides fairness."		
D — the school's data-safety lead	"Without verification rules, none of it is safe."		

The class verdict — who stays aboard, and the single best line of reasoning heard:

The point nobody made that should have been made:

WORD POWER · 19

Definition: **deepfake**

Synthetic audio or video made to look or sound like a real person.

Read it three times today — it sticks!

4.4 The Verify-It routine: four moves, no excuses

Knowledge is not defence; routine is. Here is the full analyst routine, assembled from three years of this course — four moves, in order, each cheap enough to actually run:

The Verify-It Routine — T9-18's contract

1. **TRACE** — where did this content come from? Real author, real date, real place — or a chain of reposts?
2. **TEST** — do TWO independent sources agree? If the claim has numbers, do the pieces add up?
3. **SMELL** — is it engineered to make you feel first and think later? Outrage is a distribution strategy.
4. **SEEK THE ORIGINAL** — the unedited file, the person's own account, the primary document. Deepfakes die at the original.

MY VERIFY-IT RECEIPT — ONE CLAIM FROM MY FAMILY GROUP

claim _____ h(4pt)

check the one check that decided it: _____ h(3pt)

verdict SHARE or DOUBT — and the one line I sent back: _____ h(3pt)

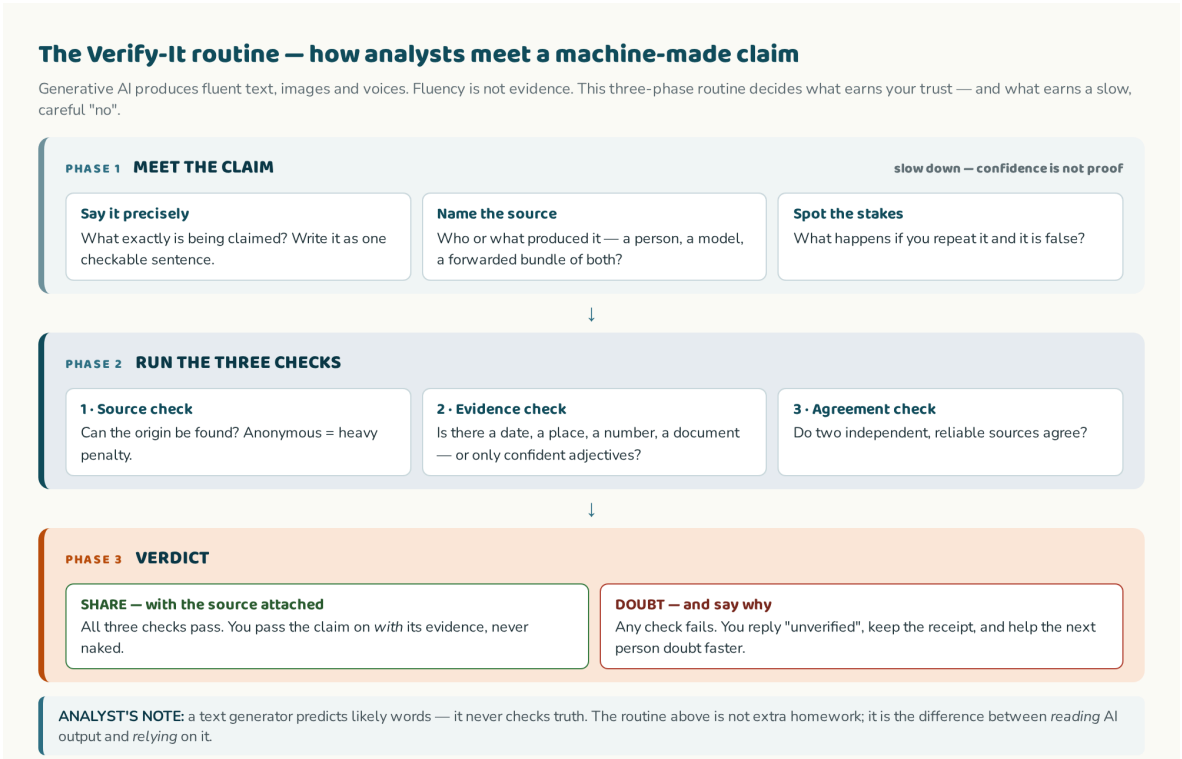


FIG The Verify-It routine as a three-phase flow: meet the claim, run the checks, deliver the verdict — with sources attached.

T9-18 Verify-It Routine

ALONE · 15 MIN

Three items arrive in your feed (invented for this mission — treat each as real). Run all four moves on paper. Commit a verdict for each: SHARE / ASK A QUESTION / DO NOT SHARE.

The item	Move 1–2 findings	Move 3–4 findings	My verdict
A forwarded clip: "The district collector announced schools close forever" — blurry, re-cropped, no date			
An AI-written "study guide" claiming the textbook's chapter 3 says something it does not			
A polished poster: "92% of parents demand phone ban" — no survey source anywhere			

Which move caught the most — and which move will you personally skip when busy? (Name it; that is the one to practise.)

WORD POWER · 20

Definition: **verification routine**

A fixed set of checks you run on claims before you trust or share them.

Read it three times today — it sticks!

MYTH BUSTER

MYTH *"Generative AI looks things up."*

TRUTH There is no library inside the machine — there are patterns, learned from text written by people. When the patterns track facts, output reads like an encyclopedia; when they don't, the same mechanism produces fluent, confident fiction, with no flag raised. That is why the machine's tone tells you NOTHING about its truth: confidence and correctness are computed separately, if the second is computed at all. You built a next-word table yourself — you watched it invent. Whatever a generator claims, your routine — trace, test, smell, seek the original — is the only lookup that actually exists.

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can explain generative AI as pattern-based prediction — not lookup — and prove it with my own next-word table ○○○
- [ ] I can name the three built-in limits — hallucination, bias, copyright — and match each to its mechanism ○○○
- [ ] I can explain what a deepfake is and why seeking the original defeats it ○○○
- [ ] I can run all four Verify-It moves on a fresh claim and commit an honest verdict ○○○
- [ ] I can argue an AI case from a stakeholder's view and score reasoning, not popularity ○○○

THINK & INK

A generated text, image or voice that fooled me (or nearly did). Now that I have built the mechanism myself, what I will do differently is …

HOME LINK *take this page home — no screen needed*

AT HOME **Family Verify-It** HOME · 15 MIN

Teach an adult the four moves, then run the routine together on one AI-flavoured item in the family phone — a forwarded video, an AI-written summary, a "study hack" post. No links to open; judge from the outside.

The item:

The move that settled it:

Our family's new rule for anything that arrives hot:

CASE NOTES — WHAT CHANGED IN MY THINKING?

Before this chapter I judged machine-made content by how it sounded. Now I judge it by …

CHAPTER 4 CLUES — POCKET THESE

- **Generative AI predicts; it never looks up** — you built the mechanism and watched it invent.
- Its three limits are built in: **hallucination, bias, copyright** — none is a removable bug.
- **Deepfakes die at the original** — seek the unedited source before the emotion picks a side.
- The Verify-It routine: **trace, test, smell, seek the original** — minutes that protect years of trust.

The clue I would tell my family tonight:

MY FEED, AUDITED — the item I will run the routine on first: Name one recent forward, post or summary in your family's phone (no links to open) and the move you predict will settle it:

Chapter 4 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

1. In your own words: why does a text generator hallucinate — and why would a bigger table not fix it completely?

2. A recruitment AI repeats ten years of one-sided hiring. Which built-in limit is this, and what data fix would you demand?

3. Name all four Verify-It moves in order — and the one move that specifically defeats deepfakes.

MY CASE JOURNAL — TODAY'S SHARPEST CLUE ABOUT MACHINES THAT CREATE AND OUR DUTY TO VERIFY

5. Capstone: The SDG-linked AI Brief

GUIDING QUESTION

Can you connect an AI idea to the world's goals — and defend it before a panel?

The final case lifts your eyes from the classroom to the world. The Sustainable Development Goals — the world's shared to-do list, from clean water to quality education — need exactly what you have practised: scoped problems, honest data, verified claims and ethical reasoning. Your team will write one SDG-linked AI brief, face an ethics review panel that argues from every stakeholder's chair, and close the year by tracing a full decision flowchart by hand — the analyst's proof that a procedure is only as good as its weakest step.

MISSIONS T9-19 – T9-21
OUTCOMES 9.R2 · 9.T1 · 9.U1 · 9.L1 · 9.D1 · 9.D2 · 9.M1 · 9.M2 · 9.M3 · 9.M4 · 9.W1 · 9.R1

STRANDS U D L W R

Links: Social Science — SDGs & development · Maths — your Chapter 3 mechanisms · English — the brief as argument

In this chapter

5.1 The brief: one goal, one data plan, one ethics note 58
5.2 The ethics review panel 60
5.3 The flowchart trace: end the year with precision 61

TERMS YOU'LL ANALYSE WITH

SDG · ethics review · flowchart · trace

WARM-UP · BEFORE YOU READ — ALONE · 2 MIN

Which ONE global problem would you fix first if you could? Write it — then be ready to defend why your choice beats your neighbour's.

ANALYST'S TOOLKIT — YOU WILL NEED: your whole notebook · the Chapter 3 mechanisms · sticky notes · a panel face

5.1 The brief: one goal, one data plan, one ethics note

The world's Sustainable Development Goals (SDGs) — clean water and sanitation, quality education, affordable clean energy, decent work, reduced inequalities — are, at bottom, **data problems**: who is missed, by how much, and what would change it. Your SDG-linked AI brief is one page of disciplined answers to six questions: WHICH goal and target, scoped with the 4Ws; WHAT data (features, sources, guest list); WHICH mechanism (your paper tools: averages, best-fit line, kNN, verification); WHAT the machine predicts; WHO decides; and the ETHICS NOTE — who could be hurt, and what the plan changes for them. One page, all evidence, no slogans.

SDG SHORTLIST — CHOOSE, THEN DEFEND THE CHOICE

The goal on our shortlist	The LOCAL version we could actually study	Why it beats the other three

The analyst's test for a good goal: can you name a FEATURE, a SOURCE and a DECIDER for it in one sentence each? If not, it is a wish, not a project.

OUR GOAL IN ONE BREATH — THE 20-SECOND PITCH

Practise on each other before the panel does: **"Our goal is …, our local problem is …, and our machine will help by …"** Two lines for the pitch that survived telling it aloud:

SDG in the wild: one place this goal already appears in the news, a government scheme, or a local poster — and the number they quote:

T9-19 SDG–AI Project Brief

GROUP · 3–4 periods MIN

One brief per team. Choose ONE SDG-linked local problem — water, education, energy, health, equality — and fill every box with evidence, not adjectives.

SECTION A — THE GOAL AND THE SCOPED PROBLEM

The SDG we chose (number + short name):

Our local version of it, scoped with the 4Ws (one sentence):

SECTION B — THE DATA PLAN

Feature (precise name + unit)	Source (who measured, how)	Guest-list check (who is missing?)

SECTION C — THE MECHANISM AND THE PREDICTION

Our paper mechanism (average / probability / best-fit line / kNN) and why it fits:

Exactly what the system outputs, and the human who decides:

SECTION D — THE ETHICS NOTE

Who could be hurt, how — and the change we made for them:

What our system will NEVER do:

5.2 The ethics review panel

Class 8's court judged a single machine; the panel judges **briefs** — and it argues in chairs. Every reviewer takes a stakeholder's seat (the served community, the data's source, the operator, the one who bears the risk) and interrogates the brief FROM that chair. The standard is the course standard: evidence for every box, a named mechanism, a filled guest list, and an ethics note that names someone specific.

T9-20 Ethics Review Panel

CLASS · 25 MIN

Each brief gets 5 minutes: 2 to present, 3 of questions from the chairs. Panelists: fill one line per chair for each brief you hear (except your own):

PANEL CARD — FOR BRIEF № _____

From the SERVED chair, the question that must be answered:

From the DATA-SOURCE chair:

From the OPERATOR chair:

From the BEARS-THE-RISK chair:

Panel shade — Emerging / Developing / Proficient / Advanced **(shade one)** ○○○○

The reverse mirror: the chair question that shook YOUR brief — and the fix version 2 adopts:

THE FOUR LEVELS — CLASS 9 EDITION

Level	What it looks like in a brief
Emerging	A goal is named; the rest is enthusiasm. The mechanism and the guest list are missing.
Developing	Real scoping and data, but the mechanism is decorative or the ethics note is a promise.
Proficient	All sections answer with evidence; the mechanism genuinely fits; the bears-ring is populated.
Advanced	Proficient, PLUS the team caught a trade-off nobody raised and defended their choice with numbers.

The bar rose with you: Class 8 rewarded honest reasoning; Class 9 rewards evidence that a sceptic can re-run.

5.3 The flowchart trace: end the year with precision

An AI system's decisions, its verify-routines, its escalation paths — all of them are **procedures**, and a procedure written well is a **flowchart**: boxes for steps, diamonds for decisions, arrows for the flow. The analyst's final skill is the **trace**: run the procedure by hand, exactly, marking the path taken and the value carried at each step. Tracing is how programmers debug, how auditors check, and how you will catch every "the machine decided" that nobody can explain.

TRACE WARM-UP — PREDICT BEFORE YOU TRACE

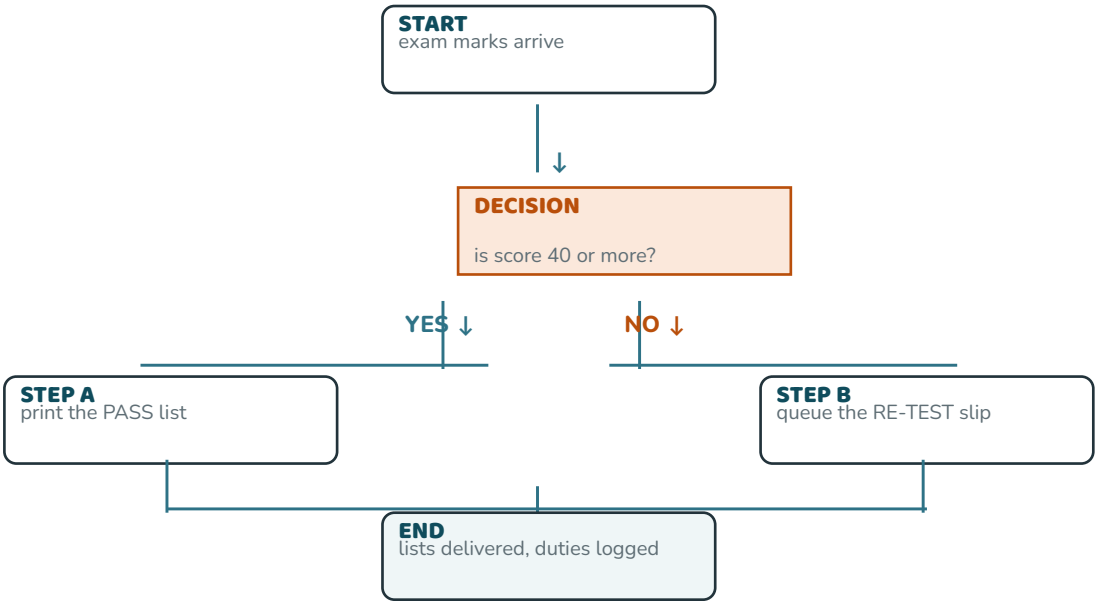
A "score 40 or more" rule sends students two ways. Which score in a class list would you check TWICE before trusting the outcome — and why that exact number?

My own procedure: a morning routine, a bus-board decision, a cricket shot selection — write one decision you make by rules, ready to flowchart it:

T9-21 Flowchart Trace

ALONE · 15 MIN

The flowchart below is the exam cell's re-test rule. Trace it for THREE students by writing the path and the final state in the table. Then improve it.



The student	Score	Path taken (boxes in order)	Final state
Meera	72		
Arun	38		
Divya	40		

The boundary case: Divya's score is exactly 40. Trace what your reading of "40 or more?" decides — and what the flowchart's writer should do to remove the doubt:

Improve it: ONE extra decision you would add (e.g., attendance below 60% → counselling first) — draw it as a box + diamond on the margin of the flowchart, and write the new rule in words:

WORD POWER · 21

Definition: flowchart

A diagram of a procedure: steps, decisions, and the arrows between them.

Read it three times today — it sticks!

AT A GLANCE · FOUR YEARS, ONE TOOLKIT

Class	You could…	The tool you trusted
6 · NOTICE	spot learning machines and ask your first questions	your senses + five questions
7 · SORT & PREDICT	train, test and watch machines get fooled	examples, hidden tests, charts
8 · BUILD THE CYCLE	plan, audit and defend an AI before it exists	the proposal + peer review
9 · REASON	compute the mechanisms and verify the claims	averages, probability, lines, kNN, Verify-It

Class 10 will ask the hardest question yet: how do we MEASURE, JUDGE and GOVERN these systems — with evidence you now know how to gather. Bring every notebook.

■ MYTH BUSTER

MYTH *"A procedure followed is a decision justified."*

TRUTH The flowchart was followed perfectly — every arrow, every box. But who wrote "40 or more"? Whose scores does the rule silently punish? A trace proves the procedure ran; it says nothing about whether the procedure is fair, current, or even sane. Machines are magnificent tracers and terrible questioners: they will follow a broken rule exactly, at scale, forever. That is why the analyst's job is never finished at "it ran correctly" — the questions **who wrote this, for whom, and who bears it** stand above every flowchart ever drawn.

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can connect a local problem to an SDG and scope it with the 4Ws ○○○
- [ ] I can write a brief whose data plan, mechanism and ethics note all carry evidence ○○○
- [ ] I can interrogate a brief from four stakeholder chairs — and score reasoning, not polish ○○○
- [ ] I can trace a flowchart exactly, spot its boundary cases, and propose a precise improvement ○○○
- [ ] I can say what my four years of AI study have equipped me to do — and what I still cannot ○○○

THINK & INK

After four case files, my honest answer to "why does AI work — and when does it fail?" is …

CASE NOTES — WHAT CHANGED IN MY THINKING?

Before this year, "AI" was a product I used. After this year, it is a set of mechanisms I can compute, question and govern. The single most useful thing I take forward is …

CHAPTER 5 CLUES — POCKET THESE

- An SDG brief is **one page of evidence**: scoped goal, data plan, mechanism, decision, ethics note.
- Panel questions come **from chairs** — the bears-the-risk chair is the one that saves the brief.
- A **flowchart trace** proves a procedure ran — never that it was fair. Boundary cases expose the writer.
- Four years, one toolkit: **notice, train, build, reason** — next stop, measure and govern.

The clue I would tell my family tonight:

MY FOUR YEARS — the mission I would teach to a Class 6 student first:

I would teach … _____ because it needs nothing but …

Chapter 5 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

1. Name the six boxes of the SDG brief — no peeking.

2. From which chair would you question a brief that has no guest-list check? What is the first question?

3. Divya scores exactly 40 and the rule says "40 or more". Why is this the flowchart's most dangerous line?

MY CASE JOURNAL — THE LAST CLUE OF CASE 4, FOR THE ROAD TO CLASS 10

ROAD TO CLASS 10 — WHAT I CARRY FORWARD

One tool from this year I will carry into Class 10 — and the one habit that makes it actually work:

My AI words

The twenty words on this case — each one met through an experience, never before.

1 problem scoping
Pinning down exactly which problem to solve, for whom, and where.
USE IT IN MY OWN SENTENCE:

2 stakeholder
Any person or group affected by a system, helpful or harmful.
USE IT IN MY OWN SENTENCE:

3 system map
A diagram naming every feature that flows in, the output, and the human decision.
USE IT IN MY OWN SENTENCE:

4 rule-based system
A program that follows if-then rules written entirely by people.
USE IT IN MY OWN SENTENCE:

5 learning-based system
A program that finds its own pattern inside examples.
USE IT IN MY OWN SENTENCE:

6 qualitative data
Data describing qualities or categories, like names and descriptions.
USE IT IN MY OWN SENTENCE:

7 quantitative data
Data measured in numbers, ready to count, average and plot.
USE IT IN MY OWN SENTENCE:

8 unstructured data
Data not in rows and columns — photos, audio, free text.
USE IT IN MY OWN SENTENCE:

9 privacy
Your control over who may know what about you, and on what terms.
USE IT IN MY OWN SENTENCE:

10 security
The locks and rules that keep data away from people who should not have it.
USE IT IN MY OWN SENTENCE:

11 mean
The sum of all values divided by how many there are.
USE IT IN MY OWN SENTENCE:

12 median
The middle value when the data is sorted in order.
USE IT IN MY OWN SENTENCE:

13 mode
The value that appears most often in the data.
USE IT IN MY OWN SENTENCE:

14 probability
Favourable outcomes divided by all equally likely outcomes.
USE IT IN MY OWN SENTENCE:

15 line of best fit
The straight line that follows a data cloud most fairly, used to predict.
USE IT IN MY OWN SENTENCE:

16 correlation
Two measures moving together — which does not prove one causes the other.
USE IT IN MY OWN SENTENCE:

17 nearest neighbour
Classifying a new point by the vote of its k most similar labelled examples.
USE IT IN MY OWN SENTENCE:

18 generative AI
Systems that create new text, images, audio or video from patterns.
USE IT IN MY OWN SENTENCE:

19 deepfake
Synthetic audio or video made to look or sound like a real person.
USE IT IN MY OWN SENTENCE:

20 verification routine
A fixed set of checks you run on claims before you trust or share them.
USE IT IN MY OWN SENTENCE:

ANALYST'S OATH

Machines **count, average, fit, vote and predict** — people **scope the problem, choose the features, read the spread, defend the k and sign the verdict**. When a number speaks, I will ask my questions: **How do I know? What is missing? Compared to WHAT? Who made this? Who is missing? What is the trick here?**

ANSWER HINTS — FOR CHECKING, NEVER FOR COPYING

T9-10: mean = 490 ÷ 9 ≈ ₹54,400 (the advertisement's number); median = ₹26,000 — the one a job-seeker deserves; no value repeats, so the honest answer for mode here is "none". Deleting 268 drops the mean to ₹27,750 — one salary moved the "average" by twenty-seven thousand. · **Name-the-trick drill:** T1, T2, T3, OK, T2. · **Data Type Sort:** voice note = QL/U (features: keywords, length); register = QT/S; slips = QL/U; readings = QT/S; photos = U (feature: intersection type); marks = QT/S.

MORE HINTS — FOR THE STICKY MISSIONS

T9-11: expect about 10 sixes and 30 odds in 60 rolls — landing within a few of that is normal, and 84 correct out of 100 for a "92%" machine is roughly three typical wobbles low: worth a second sample, not a scandal. · **Real probability drill:** 25%, 25%, 50% — and the screener: about 8 of every 26 flags are false, nearly a third, because 80 healthy people are a big pond to misfire in. Base rates rule. · **T9-12:** a fair line lands near "28 marks + 8 per hour": 3.5 hours ≈ 56; 10 hours predicts ≈ 108 — impossible on a 100-mark test, which is extrapolation confessing in public. · **T9-13:** mango 1 and 2 vote RIPE (teal majorities); mango 3 sits in no-man's land and votes NOT YET with ambers — border cases are exactly where the choice of k matters most. · **T9-16:** "The school" → team/thanked (1–1); "The monsoon" → filled/delayed (1–1); ties are honest — the machine still must pick ONE, just like you did.

MY HINT LOG — WHICH MISSIONS NEEDED A PEEK?

Hints are training data for you. Tick the mission code each time a hint unstuck you — a mission that needed three hints deserves one more retry next week, not three more hints:

- T9-__ needed a hint
- T9-__ needed a hint
- T9-__ needed a hint
- T9-__ needed a hint

A POSTCARD TO THE CLASS 10 ANALYST — WHO WILL BE YOU

The final case of the series asks: **"How do we judge, design and govern AI that affects real people?"** Class 10 measures what you now merely compute — train/test splits, confusion matrices, precision and recall, and the design of systems that deserve public trust. Leave three clues for your future self — the statistic that surprised you most, the claim you caught red-handed, and one question you dare Class 10 to answer:

My series so far — four cases, one analyst

CLASS 6 · NOTICE
I learned to **spot** learning machines and ask: how do I know?

CLASS 7 · SORT & PREDICT
I learned to **train and test** machines — and how examples fool them.

CLASS 8 · BUILD THE CYCLE
I learned to **plan, audit and defend** an AI before it is built.

CLASS 9 · REASON
I learned to **compute the mechanisms** — and verify every claim.

The habit I am proudest of now:

Where I used my questions — the tally of a finished case

My question	Where I asked it for real (at home, in class, online)
How do I know?	
What is missing? / Compared to WHAT?	
Who made this? / Who is missing?	
What is the trick here?	

Four rows is the whole method. Any row still empty is Class 10's first homework.

THE DOOR TO CLASS 10

The series finale measures what you now compute: train/test splits, confusion matrices, precision, recall and the base-rate trap — and then asks you to **design and govern** a system that deserves public trust. Keep every notebook; the definitions inside are your revision notes. And keep the six questions:

How do I know? · What is missing? · Compared to WHAT? · Who made this? · Who is missing? · What is the trick here?

THIS CASE FILE BELONGS TO THE EVIDENCE

Analyst: _____ Class & division: _____ School year: _____

Hand this book to next year's Class 9 analysts with your tallies inside — the best hand-me-down is an honest one.

ANALYST'S NOTES — anything else this case taught me that did not fit anywhere

THE CLUE I WOULD PUT IN A BOTTLE — for whoever finds this book in ten years

If a machine from 2036 reads this page, here is what I want it to know about the analysts who trained on paper:

My progress

Shade honestly — analysts never fake evidence.

MY OUTCOME TRACKER — TICK WHEN YOU TRULY CAN DO IT

- ☐ 9.U1 — tell AI, ML and DL apart; classify products as rule-based or learning-based
- ☐ 9.L1 — run the project cycle with 4Ws, stakeholders and a system map
- ☐ 9.D1 — classify data by type and structure; judge quality, sources, privacy vs security
- ☐ 9.D2 — read and critique graphs: truncated axes, cherry-picking, broken pies
- ☐ 9.M1 — compute and interpret mean, median, mode and spread; say when an average misleads
- ☐ 9.M2 — compute simple probabilities and connect them to a machine's confidence
- ☐ 9.M3 — fit a line by eye, predict, and explain why correlation is not causation
- ☐ 9.M4 — classify by nearest neighbours with distances as evidence
- ☐ 9.W1 — explain generative AI conceptually and apply a verification routine
- ☐ 9.R1 — analyse an AI case from several stakeholders' views with an ethics checklist
- ☐ 9.R2 — link an AI idea to an SDG with a data plan and an ethics note
- ☐ 9.T1 — express a decision procedure as a flowchart and trace it by hand

MY CASE FILES — SHADE A STAR PER CHAPTER FINISHED

1 Scope & Plan · 2 Data Literacy · 3 Maths Behind AI · 4 Generative AI & Ethics · 5 SDG Brief



MY OUTCOME REFLECTION — read your own ticks before you sign

The tick I am proudest of: _____ — the tick that still needs work: _____

My plan for the one that needs work (which mission I will re-run, and when):

PRATIMAI · AI HANDOUTS · CLASS 9

My AI Analyst's Certificate

This certifies that analyst

_____

ANALYST NAME

has completed the case file **"The Logic Under the Magic"** — Level 4 · REASON — computed the mathematics under the machine, briefed the SDGs, and promised to keep asking: **How do I know? · What is missing? · Compared to WHAT? · Who made this? · Who is missing? · What is the trick here?**

DATE	ANALYST'S SIGNATURE	GUIDE'S SIGNATURE
_____	_____	_____

The series concludes: Class 10 asks **"How do we judge, design and govern AI that affects real people?"** — bring every notebook you own.

My badge shelf — four years, four levels

1	2	3	4
NOTICE	**SORT & PREDICT**	**BUILD THE CYCLE**	**REASON**
CLASS 6	CLASS 7	CLASS 8	CLASS 9
badge earned ✓	*badge earned ✓*	*badge earned ✓*	*badge earned ✓*

One line per badge — what each year changed in how I think: